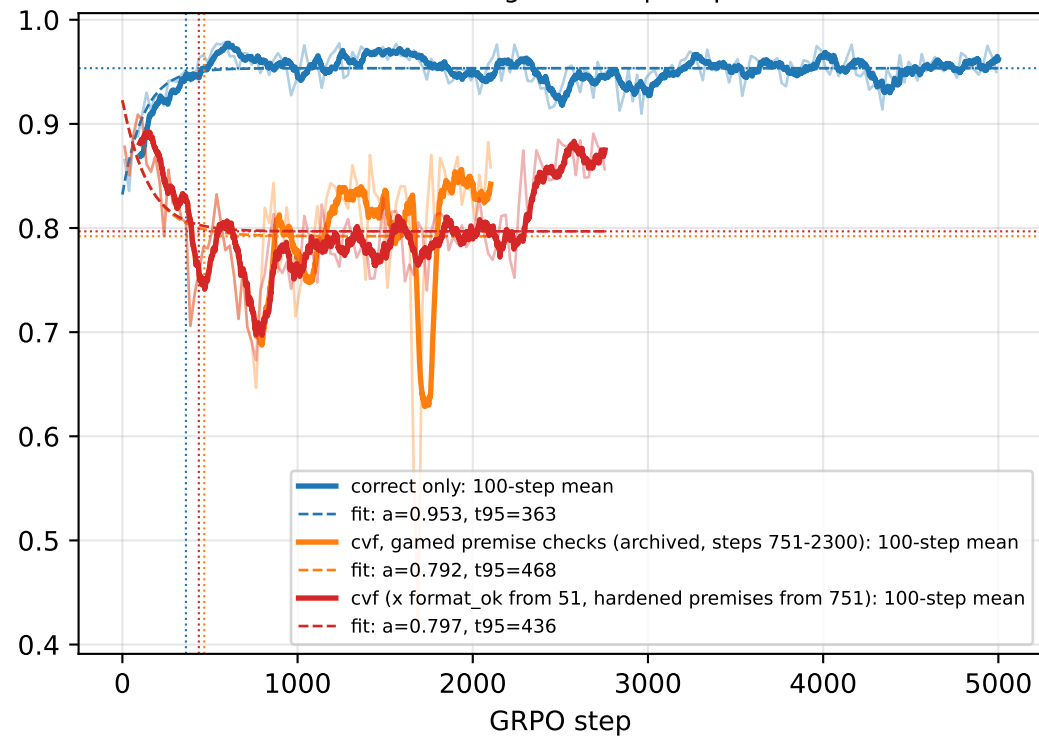


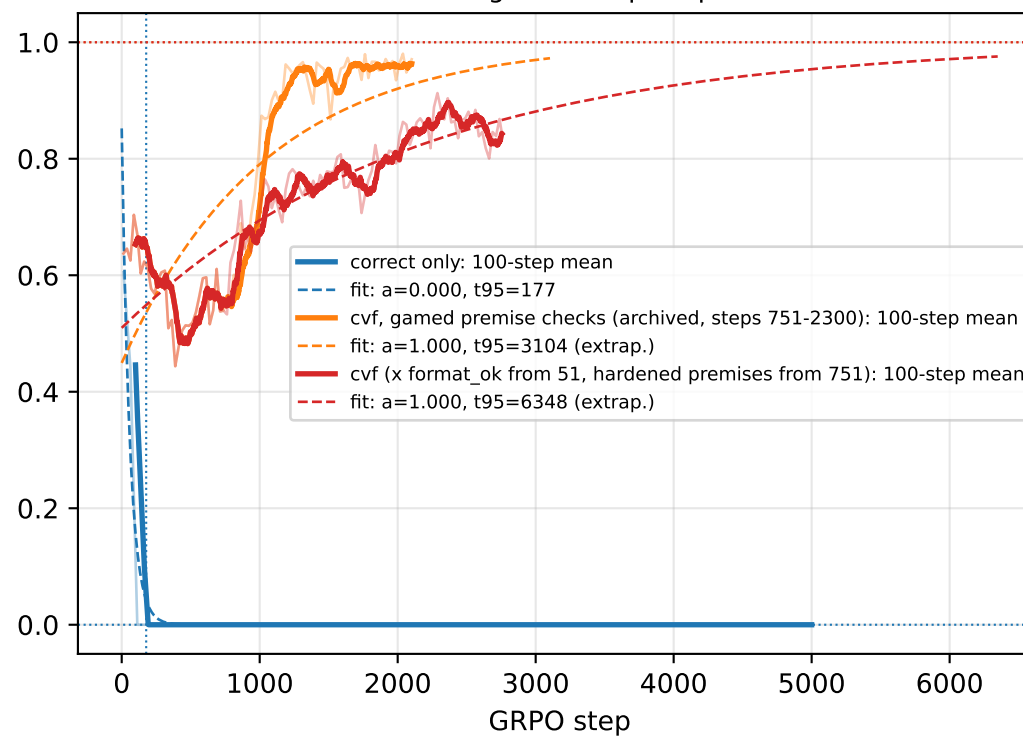
L1: GRPO reward cvf vs correct only (2B, p50 + lemma catalog SFT, T=1.0, 32x8 rollouts/step)

training-prompt rewards, 25-step means (faint), 100-step moving average; dashed: saturating exponential fit, dotted: its asymptote and t95

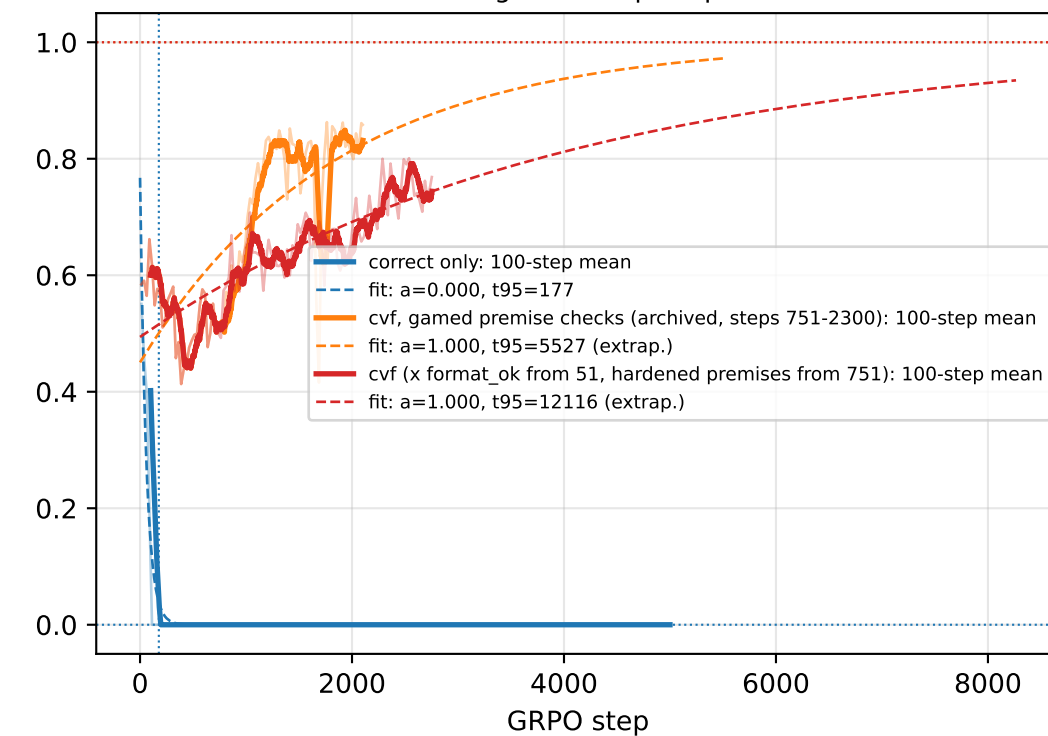
correct — generator prompts



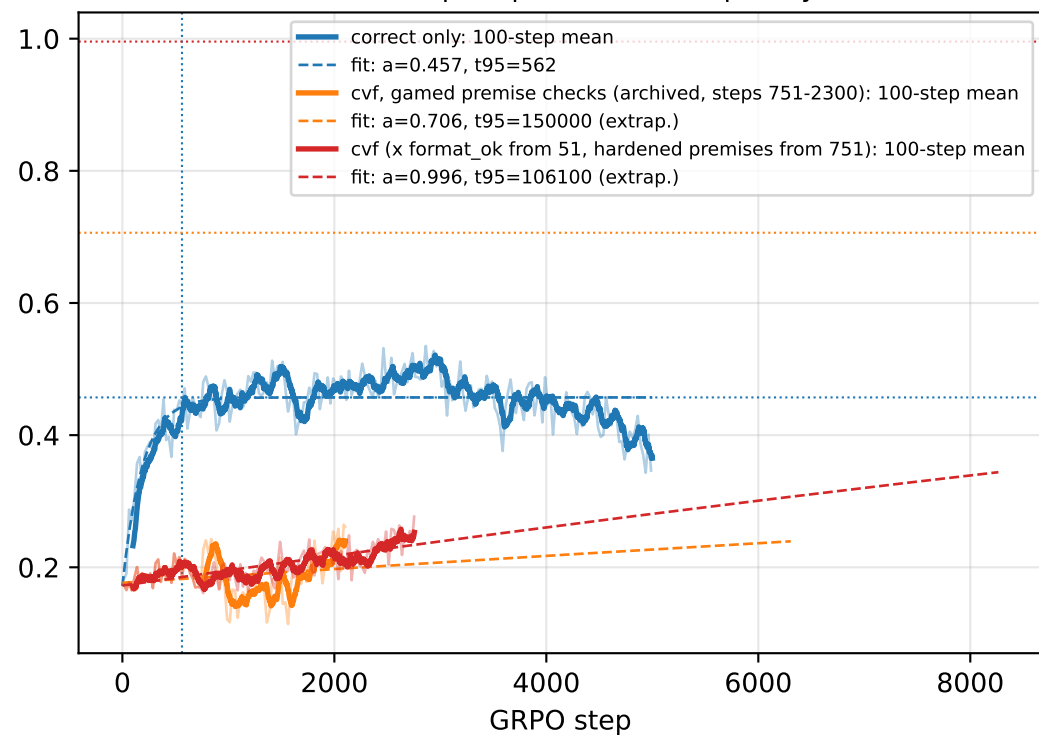
valid — generator prompts



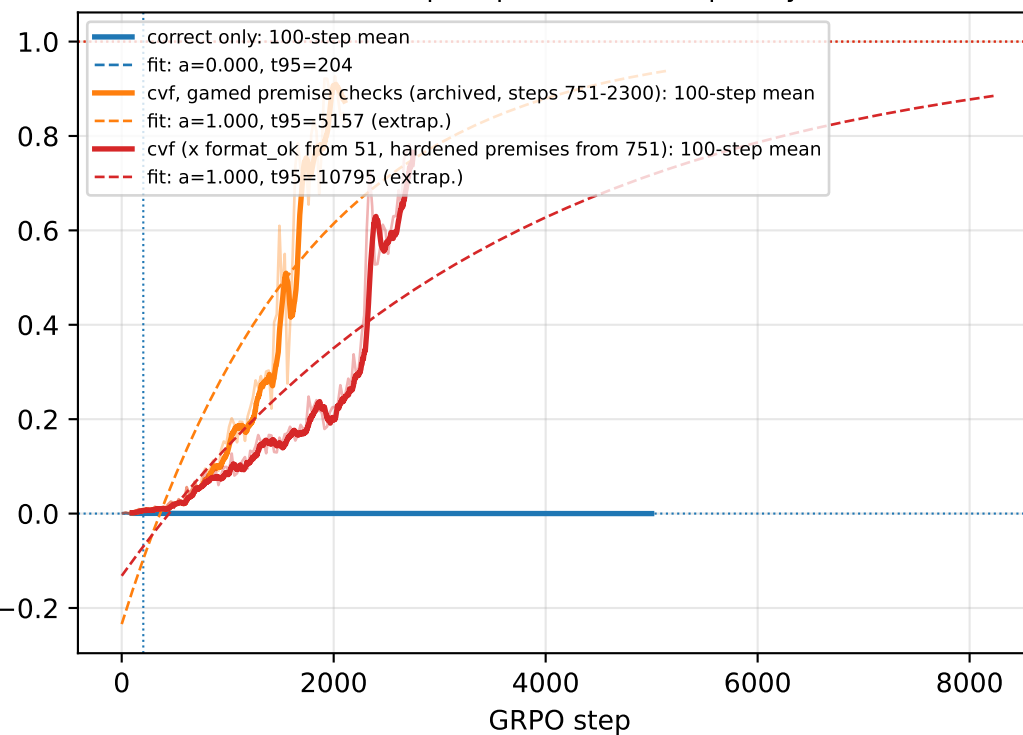
cvf — generator prompts



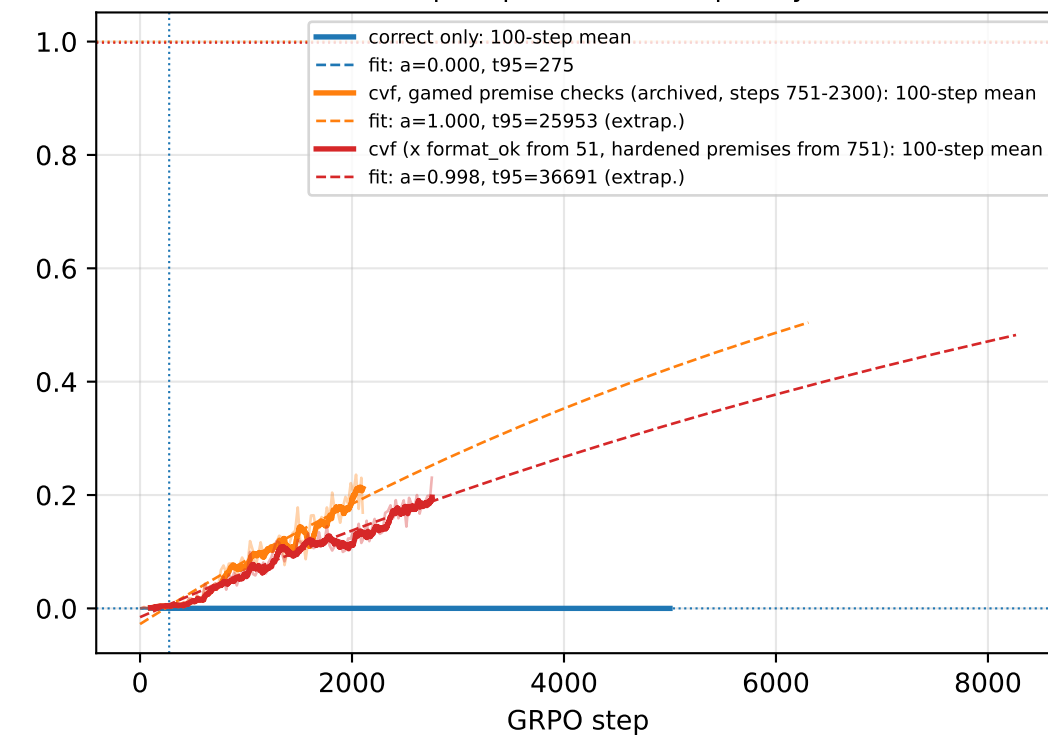
correct — Dolci prompts (math, wordprob, yesno)



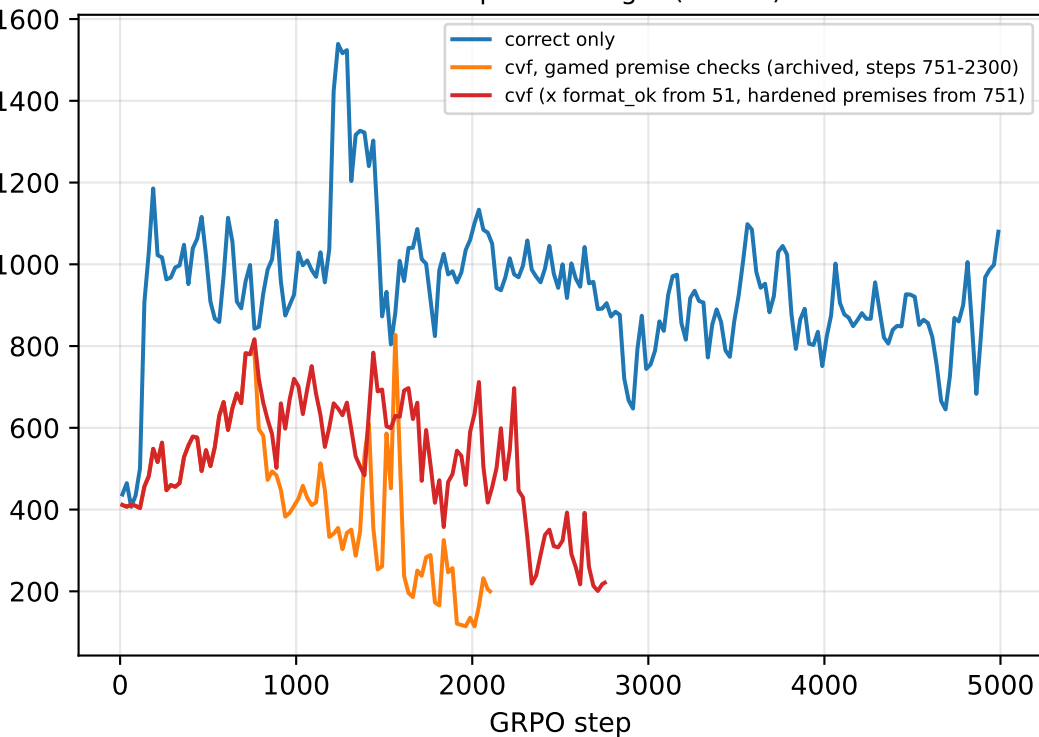
valid — Dolci prompts (math, wordprob, yesno)



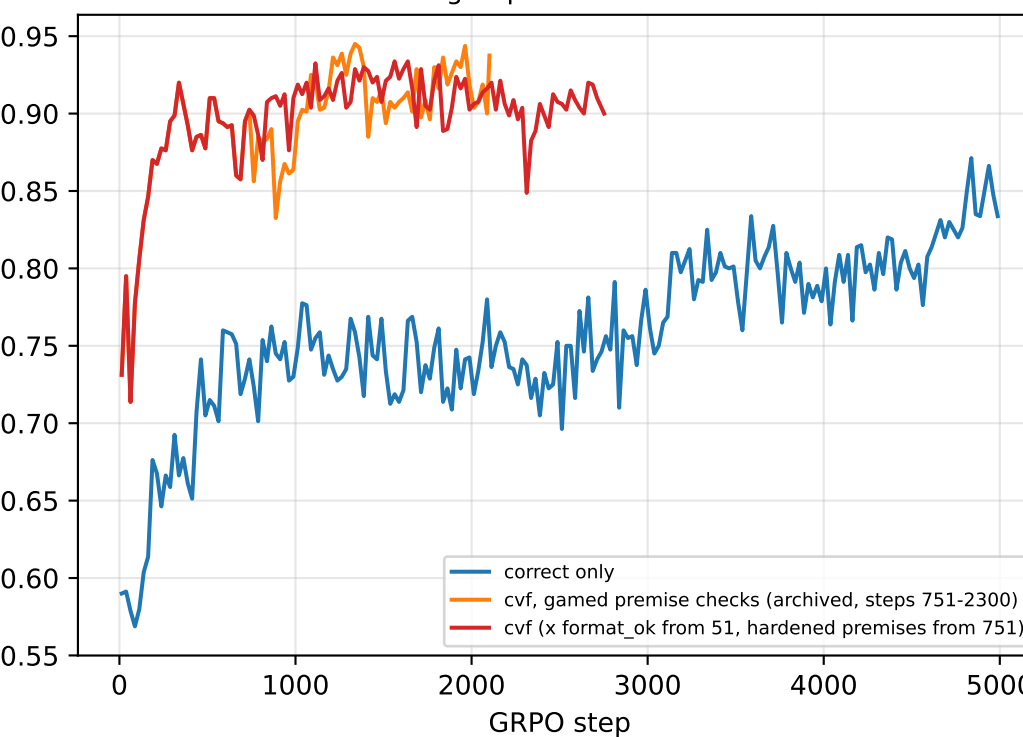
cvf — Dolci prompts (math, wordprob, yesno)



mean completion length (tokens)



fraction of groups with zero reward std



has_proof (all prompts)

